

Assessment of Data Augmentation and Transfer Learning for Making Pig Cough Classifier Robust to Changing Farm Conditions

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Abstract—Lately, the monitoring of the respiratory health status using sound signals for pig cough classification has caught the attention of the research community. In this paper, existing data augmentation techniques for Sound Event Classification (SEC) and Transfer Learning (TL) schemes are evaluated in the scenario of pig cough classification. Specifically, we talk about a) Deep Learning (DL) and TL as a popular choice for SEC and b) Enhancing model robustness using data augmentation which adds (realistic) acoustic variability to the data without requiring additional annotations. The pig cough dataset recorded in commercial farm environments is used and divided into two categories (type1 and type2) based on acoustical properties of the farms. Transfer learning with data augmentation strategies are used to explore the generalization of the pig cough classifier to changing conditions. Overall, data augmentation methods improved the performance when the model was trained on type1 data and tested on type2 data. A maximum improvement in F1-score of 3.69 percentage points and reduction of 1.66 percentage points in the F1-score standard deviation (F1-SD) was achieved with a OpenL3 based model using a hybrid data augmentation. OpenL3 based models emerge as a light weight alternative to ResNet based models for this task. In conclusion, the findings of this study call for domain specific methods to achieve classifier adaptability to changing environmental acoustical conditions in farms.

Index Terms—Deep learning, Data augmentation, Pig cough classification, Transfer learning.

I. INTRODUCTION

Precision livestock farming (PLF) is a livestock management technology that puts the livestock animals in the center and equipment around it, to measure the environment and animals' response [1]. For this purpose, a variety of sensors are used ranging from wearables (e.g., accelerometers) and contact-less sensors (e.g., cameras and microphones) to sensors that measure environmental parameters (e.g., temperature, CO₂, relative humidity, and light). In PLF, microphone-based methods are popular as they are contact-less, economically cheaper, do not require a direct line of sight, and large groups of animals can be monitored with a single sensor.

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In livestock farming, respiratory diseases lead to higher feed conversion rates and can lead to mortality when untreated. Cough is a common and most important symptom of respiratory illnesses like pneumonia, influenza, Porcine Reproductive and Respiratory Syndrome (PRRS). Continuous monitoring of sound in the farm facilitates early warning of a respiratory illness or an external factor causing the coughing. This reduces the monitoring burden of the farmer and can be used to intervene at an early stage of the illness. The advancements in computational analysis of sound scenes and events can benefit PLF significantly [2].

With the advancements of DL methods in image classification applications, there has been a growing interest to adopt it in the acoustic event classification community. Firstly, DL and TL have become extremely popular due to their performance gain and ability to learn complex patterns in the data with a limited annotated dataset. Secondly, there is a vast array of pre-trained DL based image classification models openly available in DL frameworks like PyTorch [3] and Keras [4].

Representing audio events as a spectrogram opens up the possibility to adapt and reuse the DL methods designed for image classification applications. The detection and recognition of pig wasting diseases used mel-spectrogram as the input feature representation [5]. The feature extraction and classification were achieved using Support Vector Data Description and the Sparse Representation Classifier respectively. In [6], a hybrid classifier, with a two-class Probabilistic Neural Network (PNN) and a four-class Multi Layer Perceptron (MLP), was used to distinguish coughs and metal clanging from other sounds. The work in [7] used a DL approach using Alexnet based Convolutional Neural Network (CNN) model, with spectrogram as the input data representation. The authors in [8] used a hybrid approach for feature representation by combining the features generated by CNN with hand crafted features.

With all the progress and improvements, the major challenge still lies in the model generalization to fit diverse farm conditions [9]. In the case of pig health monitoring systems, there

are two major factors influencing their performance. Firstly, the pig farm environment is harsh for cough classification systems due to the presence of high and diverse noises (e.g. ventilation and machinery). Secondly, the farm acoustics vary drastically among farms and over the growth period of the pigs. This creates undesirable variations in the perceptual and spectral characteristics of the captured sound event. The nature of the vocalizations also changes with age, breed and sex of the pigs [10]. To model these variations, there is a need to represent them in model training. Data collection for all of the discussed conditional variations is practically impossible. Moreover, top quality data annotation by trained annotators is time consuming and expensive.

This work examines the use of data augmentation methods for pig cough classification, with a focus on improving robustness against changing acoustical conditions of the farm. To the extent of our knowledge, this is the first work that compares the benefits of using data augmentation methods for pig cough classification application, on a dataset with farms belonging to two distinct acoustical types. Secondly, we look at TL by using models pre-trained on an image and audio dataset. The remainder of the paper is organized as follows. Section II describes the dataset used in the experiments. This is followed by an explanation on the Data preprocessing and Deep learning pipeline in section III. Section IV describes the different augmentation methods tested. Experimental results are discussed in section VI. Finally, we present the conclusions in section VII.

II. DATASET

A. Data Collection

The sound data used in this analysis was recorded from commercial farms. The audio data was recorded using a SoundTalks® sound monitoring device (SoundTalks N.V Belgium, 2022). This commercial product comes with a sensor system, that includes a microphone, relative humidity, and temperature sensor, and an associated web application to track the respiratory health status. During data collection the sound monitoring device was used in research mode, which enabled storing raw sound files on a hard disk attached to the data collection hardware. The data was collected in 16-bit PCM format at a sampling frequency of 22,050 Hz. The monitors were hung from the ceiling and placed at a height of 2 meters from the floor. The number of monitors placed in a compartment was decided based on the dimension of the compartment and, generally a separation of 20 meters was maintained between two monitors. The data used was from finishing pens with pigs between the age of 4 to 26 weeks. The collected data was from farms located in Europe and USA. The dataset included batches during different parts of the year.

B. Data Annotation

Preselected audio events were labeled as cough or non-cough by trained labelers. Labelers were trained with multiple expert labeled cough examples. To ensure high quality of the labels, all events were labeled by multiple labelers and labelers

were periodically evaluated. The training dataset contained data from 13 different farms, with 3000 coughs and 3000 non-cough events per farm. The test dataset was selected similarly to the training dataset, to ensure same label distribution as in the training dataset. The training set and the test set are exclusive such that no single event is part of both the training and the test set. The farm names are anonymized and renamed as farm 1, 2 and so on until 13. Depending on their farm architecture and acoustical nature, they can be broadly classified into two categories as *type1* and *type2*. The *type1* farms are smaller farms with surface area of around 500 m². The *type2* farms are large farms with surface around 900 m² and mostly have tunnel ventilation system leading to high ambient noise. Acoustically, large farms have higher reverberation effects reducing the intelligibility of the sound.

III. DATA PREPROCESSING AND DEEP LEARNING PIPELINE

The overview of the data preprocessing and DL pipeline used in this work is shown in Figure 1. The time domain audio event is augmented and transformed into a time-frequency input representation. This is followed by frequency domain augmentation layer. The Augmentation configurator controls the nature and the amount of augmentations. Next, a TL based CNN feature extractor derives meaningful feature maps. These feature maps are fed to a classifier, which is realized using a group of Fully Connected Layers (FCL). The output of the classifier is a binary class, with class 0 representing non-coughs and class 1 representing coughs.

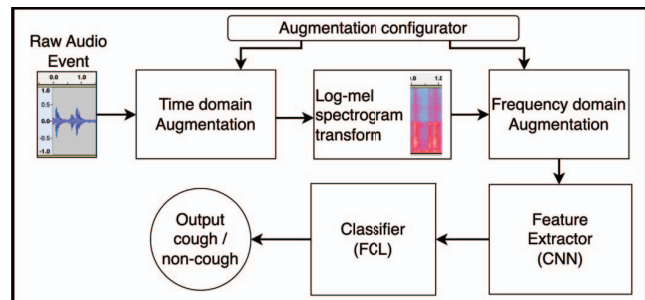


Fig. 1. Data preprocessing and Deep learning pipeline.

A. Input representation

The input representation for the DL feature extractor is the Logmel-spectrogram. Time domain audio signals are transformed to the frequency domain using Short Time Fourier Transform (STFT). Subsequently, it is converted by a number of mel filter banks to a lower dimensional mel representation [11] (see Figure 2). Mel is a perceptual scale for the audio frequencies. It is transformed to decibels, which is a closer representation of how sound intensities are perceived by human ear. Logmel spectral features are a popular choice for input representation in DL based acoustic event classification tasks [12].

Individual audio events of 1 second duration were converted from time domain audio signal to mel-spectrogram using the

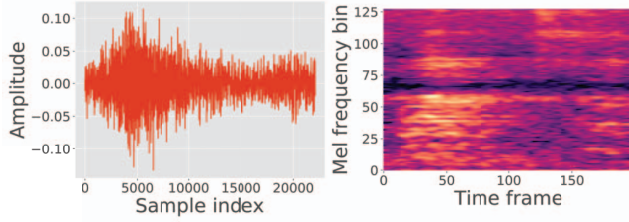


Fig. 2. Time domain (left) and Frequency domain (right) representation of a pig cough event.

melspectrogram function and the conversion to decibel scale was done using the *power_to_db* function from the librosa library [13]. The lowest frequency to be considered to derive the mel spectral bins was set to 50 Hz to avoid very low frequency bias due to the ventilation noise in the farms. STFT window length was set to 2048 with a hop length of 111 samples. The number of mel bins was set to 128. Each event is transformed to a Logmel-spectrogram of size 199×128 , where 199 represents the number of time frames and 128 corresponds to the number of Log-mel frequency bins. The configuration of the input representation was set to match that of the OpenL3 based models.

B. Transfer Learning

TL is the method of learning feature representations in a source domain and transferring this knowledge to the target domain. The models pre-trained on source domain data are fine-tuned with the limited target domain data to learn an efficient model for the target task. Numerous studies [7], [8] have shown to benefit from TL with image as the source domain and audio as the target domain. TL approach is used here by pre-training the model on Imagenet dataset [14]. In this work, we used ResNet50 and OpenL3 based pre-trained models.

1) *ResNet based models*: The concept of residual training was first presented in [15]. ResNet architecture was proposed for image classification tasks in ILSVRC [14] & COCO 2015 [16] competitions and showed advantages in both complexity and performance compared to well-known DL model like VGG [17]. The presence of skip connections makes the ResNet architecture resilient against vanishing gradients and helps the network layers to learn the identity functions efficiently. The skip connections feed the output of a layer to multiple subsequent layers. ResNets work by learning the residual functions with respect to the inputs at a given layer. The ResNet50 architecture has a layer depth of 50. Among the popular Imagenet pre-trained models like VGG, AlexNet, DenseNet and GoogleNet, ResNets seems to perform better in SEC tasks [18]. For ResNet50 based models, input mel-spectrogram is repeated 3 times to obtain a 3 channel input, making it compatible with Imagenet pre-trained model. The fully connected layers of the ResNet50 are replaced by a shallow FCL of 16 neurons followed by an output layer of 1 neuron corresponding to the binary output class. The fully

connected layer is trained along with the fine tuning of the CNN layers using the pig cough dataset.

2) *OpenL3 based models*: OpenL3 is an audio embedding framework developed using the L3-Net proposed in Look, Listen, and Learn concept presented in [19]. L3-Net has audio and video sub-networks trained jointly using audio and video data in an unsupervised fashion. The pre-trained audio sub-network is used in OpenL3 to provide quality audio embeddings. Experiments in [19] have shown that OpenL3 variants outperforms VGGish and SoundNet in SEC tasks. OpenL3 is a pre-trained network specifically for audio data, unlike the Imagenet pre-trained models that are trained on image data. The OpenL3 feature extractor outputs an embedding of size 6144 and the FCL consists of 3 layers of size 128,64 and 16 respectively. An Output neuron of size 1 makes the binary prediction for coughs.

IV. DATA AUGMENTATION

Data augmentation is the process of creating new data from an existing dataset using different transformations. It aims to synthetically create the variations in the training dataset, which is representative of the vast variation present in real-life situations. It was proven to be effective to prevent overfitting [20]. Some of the commonly used augmentation techniques in image classification tasks include rotation, shear, height shift, width shift, brightness changes, reflection, and zooming. Although there are many data augmentation transformations to generate synthetic data, the validity of the transformations depends on the domain of the data and the application. In case of SEC, certain transformations like reflection are not a valid augmentation method. This is because, these transformations completely defy the fundamental characteristics of the spectral dynamics of the event. In audio spectrograms, which are time-frequency maps, the order of the sequences along the time dimension is fundamental to the nature of the event. Performance can degrade severely when certain inappropriate transforms are used in data augmentation [21]. Specifically for the audio data, the literature reveals augmentation approaches that operates in the time and the frequency domain. These are elaborated in the next subsections. Also, the time and the frequency domain augmentation methods can be applied serially to create an augmentation chain.

A. Time domain audio augmentation

Time domain or raw audio data augmentation refers to the time domain signal transformations like background noise addition, distortions, gain, pitch shifting, loudness compensation, normalization, time stretching and Room Impulse Response (RIR) convolution. In this work, an Audiomentations [22] augmentation chain, including pitch shifting, noise addition, and RIR convolution was used. The range of pitch shifting was set to 1 semitone in either direction. The noise samples used were collected from real farm conditions, which consisted of ventilation and machinery noises. Different levels of noise signal (with scaling factor between 0 and 3) were added to the original files to make them realistic representation of the

noisy cases encountered in pig barns. The noise used in this study focuses mainly on diffused noise sounds.

The Ecotheif RIRs dataset [23] contains 115 RIRs, covering a wide range of rooms with estimated RT60 values ranging from 0.07 s to 4.2 s, was used for RIR augmentation. Understanding the fact that the original audio files are not very clean and contains the room effects of the farm, the RIRs from Ecotheif dataset were chosen to add enough reverb while not making the audio look unnatural. To achieve this, we selected 51 RIRs with RT60 values smaller than 1. The RT60 values were measured using the pyroomacoustics library [24].

B. Frequency domain data augmentation

Frequency domain augmentation works on the frequency domain representation which is mel-spectrogram in our case. The methods tested include SpecAugment, MixUp and CutMix.

1) *SpecAugment*: SpecAugment was proposed in [25] and was applied in the context of speech recognition. The augmentation process includes warping followed by time and frequency masking. Warping deforms the signal in time axis and gives twisting or bending effect around a selected point [25]. Time warping was used in [26] to augment the time series data. Time frequency masking performs random deletion in frequency and time axis of the spectrograms and the deleted bands are filled in to generate augmented audio spectrograms. The masked portions were replaced with zeros, which was also the mean of the normalized data. The parameters of the SpecAugment include the warping parameter, the number and the dimensions of the masking bands. The number of masking bands was set to 3 for both time and frequency axes. The range of frequency masking was set from 2% to 4% and the range of time masking was set between 1% and 2%.

2) *MixUp*: MixUp was proposed in [27], where the newly created examples are a convex combination of pairs of existing data samples (see Equation 1).

$$\begin{aligned}\tilde{x} &= \lambda x_i + (1 - \lambda)x_j \\ \tilde{y} &= \lambda y_i + (1 - \lambda)y_j\end{aligned}\quad (1)$$

Where, x_i, x_j are raw input vectors and y_i, y_j are corresponding one-hot label encoding. x_i, y_i and x_j, y_j are two examples drawn at random from the training data. λ is sampled from a Beta distribution function with alpha and beta values of 0.2 and $\lambda \in [0, 1]$. Experiments have shown that MixUp encourages the training process to generate a linear behaving classifier, for predicting in-between examples seen during training and thereby improving its stability [27].

3) *CutMix*: CutMix augmentation was proposed in [28], where patches of images were cut and inserted into other training images. The corresponding labels are also mixed in proportion to the area of the inserted patches (see Equation 2). The newly created synthetic training examples encourage the learning process to learn localizable features and thus efficiently use the training pixels. CutMix was seen to outperform contemporary augmentation techniques for image classification tasks. Both CutMix and MixUp methods use the

full image region and mixes both the images and labels, and uses regional dropout.

$$\begin{aligned}\tilde{x} &= \mathbf{M} \odot x_A + (1 - \mathbf{M}) \odot x_B \\ \tilde{y} &= \lambda y_A + (1 - \lambda)y_B\end{aligned}\quad (2)$$

Where, x_A, x_B are raw input vectors and y_A, y_B are corresponding one-hot label encodings. x_A, y_A and x_B, y_B are two examples drawn at random from the training data. λ is sampled from a Beta distribution function with alpha and beta values of 0.2 and $\lambda \in [0.2, 0.8]$. $\mathbf{M} \in \{0, 1\}^{W \times H}$ denotes a binary mask indicating where to drop out and fill in from two images, and $\odot$ is element-wise multiplication. λ decides the proportion of mixing and accordingly the region of image cut was randomly selected from the possible regions.

V. EVALUATION CRITERIA

The performance of the audio classification system is measured in terms of accuracy, precision, recall and F1-score. These metrics are derived from the values of True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN) (see Equation 3). For comparing the methods, we decided to use F1-score as the metric, which jointly represents both Precision and Recall of the classifier system in a single metric. The standard deviation of the F1-score among farms (F1-SD) is also measured.

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (3)$$

VI. EXPERIMENTAL SETUP AND RESULTS

Firstly, we experimented with different data augmentation methods, by training ResNet50 based feature extraction models (see Section III) with a Stochastic Gradient Decent (SGD) optimizer with an initial learning rate of 1e-2. The training was done on mini batches of size 128 for 50 epochs, which was sufficient for the model to achieve convergence. Binary cross entropy was used as the loss function. The second set of experiments were done with OpenL3 based models for different data augmentation methods. OpenL3 based models were also trained with Stochastic Gradient Decent (SGD) optimizer with an initial learning rate of 1e-2. The training is done on mini batches of size 128. The model is trained for 50 epochs, which was sufficient for the model to achieve convergence. Binary cross entropy is used as the loss function. The training was done on a server with AMD EPYC 7402P processor and NVIDIA GeForce RTX 2080 Ti GPUs. The models were implemented in Python3 using Keras DL framework [4]. Performance evaluation was done on the test set, described in section II-B.

A. Baseline performance

To establish the baseline, we generated different variants of the ResNet50 based model, depending on the farms included in the training data. In Figure 3, *all,type1* and *type2* refer to the models trained with the data from all farms, *type1* farms and *type2* farms respectively. The performance of the

model *all* is consistent across farms and is considered as the target performance in the rest of the paper. However, the performance drops significantly for the farms which were not included in training, in comparison to the farms that were part of the training dataset. The drop in performance on data from an unseen farmtype is more pronounced when the model was trained with *type1* data. This is mainly due to nature of the *type2* farms (see Section II-B) and hence it encompasses some of the characteristics of the *type1* farm data. Data augmentation on *type2* data to make the model generalize to *type1* farms is tricky as it will require complex methods like dereverberation and denoising to go back to the data observed in *type1* farms. Hence, in the following experiments we will evaluate the transfer learning and data augmentation methods for the generalization of the model, by training the model with *type1* data and validate the performance on *type2* data.

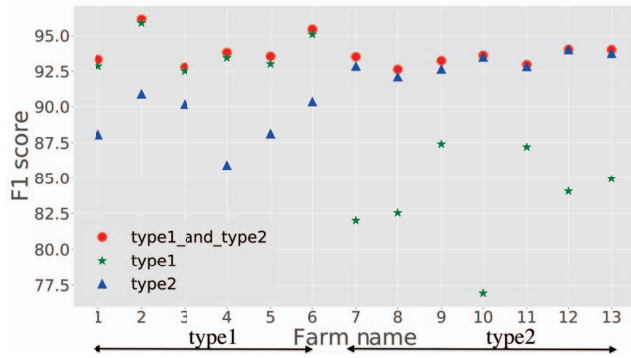


Fig. 3. ResNet50 based model performance across farms.

B. Comparison of data augmentation techniques

Table I states the F1-score values for different farm types as well as the training dataset size while using different data augmentation methods on the training data from *type1* data. Hybrid is an augmentation scheme formed by the combination of Time domain augmentation and Frequency domain augmentation (MixUp, and SpecAugment). The Hybrid augmentation scheme tries to encompass larger possible variations by combining different augmentations. ResNet50 based model shows performance drop in *type2* farms, when the model was trained only using *type1* farm data. CutMix augmentation does not improve the performance but rather lowers it. This could be due to the fact that the cut portion from a cough event can be the frames after the cough part has ended in the one second event window. Even though the cough events are represented as one second audio events, the actual portion of the event containing the cough part could vary depending on the factors of the underlying biological system. MixUp and SpecAug marginally improved the F1 score for both *type1* and *type2* farms. RawAudioAug gave the best improvement of 2.54 percentage points in F1 score and reduced the F1-SD (see Section V) by 1.54 percentage points. Hybrid augmentation resulted in similar performance as with RawAudioAug. With OpenL3 based model, Hybrid augmentation improved the

TABLE I
PERFORMANCE TABLE FOR DATA AUGMENTATION METHODS WITH MODELS TRAINED ON *type1* DATA.

Augmentation	F1 type1	F1 type2	F1-SD	Training data size
ResNet50 None	93.80	83.58	5.71	1x
ResNet50 CutMix	93.94	81.58	6.99	2x
ResNet50 MixUp	94.16	84.11	5.55	2x
ResNet50 SpecAug	94.32	84.32	5.54	2x
ResNet50 RawAudioAug	94.26	86.79	4.17	4x
ResNet50 Hybrid	94.33	86.12	4.51	4x
OpenL3 None	93.52	82.87	6.13	1x
OpenL3 RawAudioAug	93.72	86.12	4.49	4x
OpenL3 Hybrid	94.03	86.56	4.47	4x

type2 farm F1 score by 3.69 percentage points and the F1-SD reduced by 1.66 percentage points. OpenL3 based model with augmentation gave similar performance as the ResNet50 based model, which makes it a light weight alternative for ResNet50 based model.

C. Effectiveness of Data Augmentation methods with changing training dataset size

In this experiment, we investigated the effectiveness of the data augmentation method while varying the size of the original training dataset. We used the *OpenL3 Hybrid* configuration mentioned in Table I. We trained different variants of the *OpenL3 None* and *OpenL3 Hybrid* configuration by changing the size of the training dataset. Figure 4 shows the performance graphs as well the relative boost in performance with augmentations with changing size of the original training dataset. The improvement is higher at lower training dataset size with a peak boost of 5.88 percentage points at 10% training size and gradually decreases with increasing training dataset size to obtain a boost of 3.17 percentage points at 100% training size. Higher the amount of training samples, the coverage of the data distribution increased and thereby reducing the impact from the data augmentation methods.

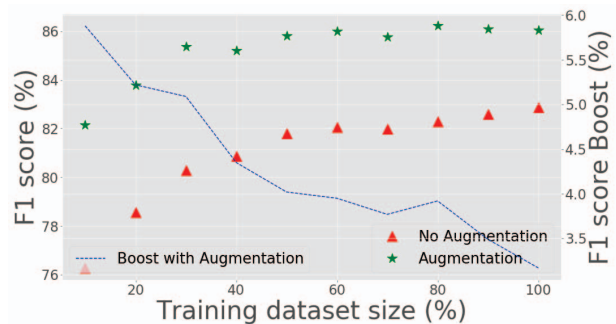


Fig. 4. Performance with varying training data size.

VII. DISCUSSION AND CONCLUSION

The presence of a representative dataset is extremely important in building data driven DL models for pig cough classification, where the feature representations are derived from the data. We show that the models trained on *type1* farms do not generalize well to *type2* farms where the latter has much larger room dimensions and ambient noise. Transfer Learning with data augmentation improves the mean performance by 3.69 percentage points as well as reduces F1-SD by 1.66 percentage points. OpenL3 based models with 5.6 million trainable parameters, emerges as a lightweight alternative to ResNet50 based models with 23.6 million trainable parameters, achieving similar performance.

Out of the different augmentation methods tested, Hybrid augmentation scheme gave the best improvement in performance. The noise used in this study mostly included diffused noise sources. However, Directional and interfering noisy events like knocks are also important in this context and will be explored in future work. Image inspired augmentation methods like CutMix and MixUp, did not show promising improvements. Combining different augmentation methods can give better performance compared to a single augmentation strategy. The data augmentation boosts the performance significantly at lower training dataset sizes and gradually reduces the impact as training size increases. Though there are obvious gains in using data augmentation, there is a considerable gap in performance of *type2* farms when trained on *type1* farms. To bridge this gap, there is a need to explore approaches using domain specific knowledge of the environment and utilize it to make more intelligent classification.

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